Predictive CI/CD Reliability

Predicting Workflow Failures from Pre-Build Metadata

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Degree Program: MTech - Integrated Software Systems - WIPRO TECHNOLOGIES (WASE)

Research Area: Machine Learning

Dissertation carried out at: Wipro Technologies, Pune



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1. Broad Area of Work

Continuous Integration and Continuous Delivery (CI/CD) pipelines are a core part of modern software engineering, and predicting their outcomes is a growing research area in machine learning for software engineering. This project predicts whether a CI workflow run will fail, using only information available before the run starts. It covers the following application areas:

- Software engineering analytics: To study CI workflow histories, commits and repository metadata from public projects
- Supervised machine learning: To predict workflow failure with Logistic Regression, Random Forest and gradient boosting, against strong baselines
- Temporal data engineering: To build leakage-safe historical features that never use information from after the prediction time
- Evaluation and MLOps: To evaluate on future builds and unseen repositories, calibrate risk scores and deliver an inspectable service

2. Background

Teams that practice CI trigger automated build and test workflows on every commit. When a workflow fails, developers wait longer for feedback, compute resources are consumed and work is blocked until the cause is found (Hilton et al., 2016). If the risk of failure could be estimated before a workflow runs, reviewers and pipeline owners could direct attention to the riskiest runs first. This is relevant to any organization that depends on automated pipelines for delivery, where late detected failures slow releases and increase cost.

Earlier studies have predicted CI build outcomes with machine learning (Hassan and Wang, 2017; Ni and Li, 2017; Saidani et al., 2020). However, results are often hard to trust. Random train/test splits mix past and future builds, some features are only known after a build starts, and models are rarely compared with a simple baseline such as each repository's own historical failure rate. Failures also have many causes (code regressions, flaky tests, infrastructure and configuration problems), so a prediction is not a statement about code quality.

This project addresses these gaps by fixing its evaluation rules before testing: a go/no-go data audit, a strict prediction-time definition, time-based and unseen-repository holdouts, and an acceptance criterion measured against the historical-rate baseline and the best possible ranking, with confidence intervals.

3. Objectives

The objectives of my project are as follows:

Audit a public CI dataset (outcome labels, timestamps, identifiers, feature availability) and apply a written go/no-go gate before any modelling.
Define a prediction contract and build a leakage-safe feature pipeline in which every run uses only information available at its prediction time, verified by automated tests.
Compare a dummy baseline, Logistic Regression, Random Forest and HistGradientBoosting with a smoothed historical repository failure-rate baseline, using Average Precision and recall at a 10% alert budget.
Evaluate generalization on future builds and on repositories unseen in training, with calibration, feature-group ablation and error analysis.
Deliver a FastAPI service and a two-page Streamlit dashboard for inspecting historical builds, together with a model card that states the limitations.

4. Scope of Work

The student will independently carry out the dataset audit, feature engineering, model development, evaluation, and the prediction service and dashboard. The target is workflow-level failure (success versus failure) predicted from pre-build metadata, with repository history reconstructed only from runs that had already finished.

Out of scope: live integration with GitHub, identifying the cause of a failure or labelling code as defective, automatic build cancellation, deep-learning models, CatBoost/LightGBM, and free-form what-if predictions.

Success is judged against a criterion fixed in advance: recall at the 10% alert budget of at least 0.20 and compared with the historical-rate baseline using repository-clustered bootstrap confidence intervals. A missed target will be reported honestly as a finding.

5. Plan of Work

Phases	Start Date-End Date	Work to be done
Data Audit & Go/No-Go	02 Nov 2026 – 08 Nov 2026	Dataset selection, data dictionary, label and timestamp audit, baseline feasibility, go/no-go decision
EDA & Feature Engineering	09 Nov 2026 – 22 Nov 2026	EDA, target definition, frozen data splits, leakage-safe features with automated tests
Modelling & Tuning	23 Nov 2026 – 06 Dec 2026	Baselines, Logistic Regression, Random Forest, gradient boosting, tuning, class-weight and buffer decisions
Evaluation	07 Dec 2026 – 20 Dec 2026	Calibration, thresholds, single final test run, temporal and unseen-repository evaluation
Analysis & Model Card	21 Dec 2026 – 27 Dec 2027	Feature-group ablation, error analysis, feature importance, model card
Application Development	28 Dec 2027 – 14 Jan 2027	FastAPI service, two-page dashboard, Docker packaging and automated CI
Dissertation Review	15 Jan 2027 – 20 Jan 2027	Contingency buffer; submit Dissertation to Supervisor & Additional Examiner for review and feedback
Submission	21 Jan 2027 – 30 Jan 2027	Final review and submission of Dissertation

6. Literature References

The state of the art is the base for any successful research project. The current project draws on literature about CI build-outcome prediction, flaky tests, probability calibration and evaluation on imbalanced data. The following are referred works from the preliminary literature review.

1. M. Beller, G. Gousios, and A. Zaidman. TravisTorrent: Synthesizing Travis CI and GitHub for full-stack research on continuous integration. In Proceedings of the 14th International Conference on Mining Software Repositories (MSR 2017), pages 447–450. IEEE, 2017.
2. F. Hassan and X. Wang. Change-aware build prediction model for stall avoidance in continuous integration. In Proceedings of the 11th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM 2017), pages 157–162. IEEE, 2017.
3. M. Hilton, T. Tunnell, K. Huang, D. Marinov, and D. Dig. Usage, costs, and benefits of continuous integration in open-source projects. In Proceedings of the 31st IEEE/ACM International Conference on Automated Software Engineering (ASE 2016), pages 426–437. ACM, 2016.
4. Q. Luo, F. Hariri, L. Eloussi, and D. Marinov. An empirical analysis of flaky tests. In Proceedings of the 22nd ACM SIGSOFT International Symposium on Foundations of Software Engineering (FSE 2014), pages 643–653. ACM, 2014.

5. M. Mitchell et al. Model cards for model reporting. In Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* 2019), pages 220–229. ACM, 2019.
6. Z. Ni and B. Li. Cost-effective build outcome prediction using cascaded classifiers. In Proceedings of the 14th International Conference on Mining Software Repositories (MSR 2017). IEEE, 2017.
7. A. Niculescu-Mizil and R. Caruana. Predicting good probabilities with supervised learning. In Proceedings of the 22nd International Conference on Machine Learning (ICML 2005), pages 625–632. ACM, 2005.
8. F. Pedregosa et al. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12:2825–2830, 2011.
9. I. Saidani, A. Ouni, M. Chouchen, and M. W. Mkaouer. Predicting continuous integration build failures using evolutionary search. Information and Software Technology, 128:106392, 2020.
10. T. Saito and M. Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLoS ONE, 10(3):e0118432, 2015.

7. Particulars of the Supervisor and Examiner

Particular	Supervisor	Additional Examiner
Name	Apurva Deshpande	Ram Panchal
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8. Remarks of the Supervisor

The proposed work, *Predictive CI/CD Reliability: Predicting Workflow Failures from Pre-Build Metadata*, addresses a practical problem in software delivery: failed CI workflows cost compute and delay feedback to developers. The topic suits a dissertation in machine learning for software engineering.

The objectives are clear and achievable within the semester. The proposal is careful about evaluation. It includes a data audit with a go/no-go decision before modelling, a strict rule on what information is available at prediction time, and features built so they don't leak later information. It tests on later builds and on repositories the model hasn't seen. The success criterion is fixed in advance and compared against each repository's historical failure rate. These choices avoid the most common weaknesses in earlier build-prediction studies.

The scope is realistic, and the plan correctly excludes finding the cause of a failure and connecting to GitHub live. The main risk is the data: a suitable public CI dataset with reliable timestamps and labels must be available. The planned audit addresses this. I recommend the student record the go/no-go decision and its reasoning in the dissertation.

The timeline is reasonable and includes a contingency buffer. I recommend that the outline be approved.

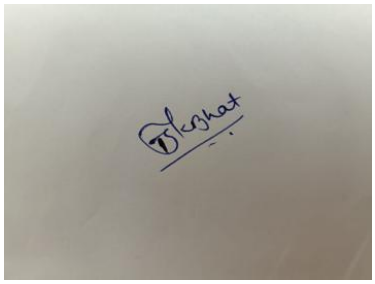
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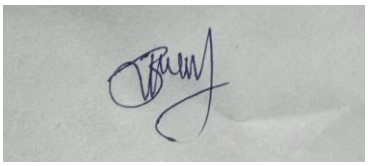
First Semester of Academic Year 2026-2027

(Insert Course No.) : Dissertation / Project / Project Work Outline

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Signature of Student	**Signature of Supervisor**	**Signature of Additional Examiner**
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